

Winding number on 3D lattice

Okuto Morikawa

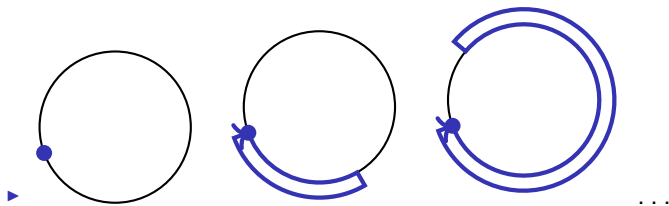
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2/5/2025 CREST Seminar

- OM, H. Suzuki, [[arXiv:2412.03888 \[hep-lat\]](#)].

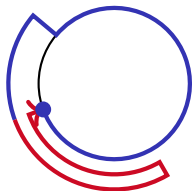
Topological number

- Mapping S^1 to $U(1)$:



- ▶ Identical under any continuous deformation, **homotopic**.

- Nontrivial (not homotopic) case:



The number of times to wrap
= **topological number (winding number)**

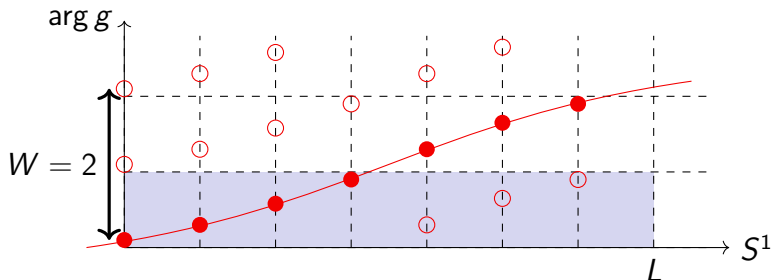
- Nicely defined for continuous maps (math)

Topological number (discrete case)

- Many topological quantum numbers to characterize phase transitions in physics.
- In practice, difficult to consider
 - ① infinite-dimensional function space,
 - ② systems with infinite degrees of freedom like QFT, ...

Then, some discrete techniques are used

- ① truncation of Hamiltonian (Hilbert space),
 - ② discretized Brillouin zone, (lattice) regularization, ...
- E.g., $g : S^1 \rightarrow U(1)$ is replaced by $g_L : \mathbb{Z}/L\mathbb{Z} \rightarrow U(1)$



Topological charge in lattice Abelian gauge theory

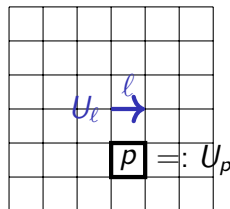
- Lattice $U(1)$ gauge theory

$$U_\ell \in U(1) \quad \text{at link (bond)} \ell = (x, \mu),$$

$$U_p \equiv U_{x,\mu} U_{x+\hat{\mu},\nu} U_{x+\hat{\nu},\mu}^\dagger U_{x,\nu}^\dagger \quad \text{at plaquette } p = (x, \mu, \nu).$$

- 1st Chern number

- ▶ $A_\ell \equiv \frac{1}{i} \ln U_\ell, \quad -\pi < A_\ell \leq \pi.$
- ▶ $F_p \equiv \frac{1}{i} \ln U_p, \quad -\pi < F_p \leq \pi.$
- ▶ $F_{\mu\nu} = \Delta_\mu A_\nu - \Delta_\nu A_\mu + 2\pi N_{\mu\nu},$
 Δ_μ : forward difference op.,
 $N_p \in \mathbb{Z}$ s.t. $-\pi < F_p \leq \pi.$
- ▶ Lattice Chern number on T^2



$$c_1 = \frac{1}{2\pi} \sum_x F_{12}(x) = \sum_x N_{12}(x) \in \mathbb{Z}$$

[Fujiwara–Suzuki–Wu '00, Fukui–Hatsugai–Suzuki '05]

- 2D case: well-established discrete formula of Chern number to reproduce a quantized value

Discrete formula of winding number

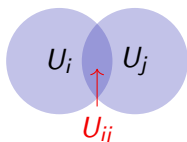
- General cases: difficult to reproduce topological numbers
 - ▶ Input discrete data $\longrightarrow$ Output $W \in \mathbb{Z}$?
- Proof of topological charge for non-Abelian gauge theory
 - ▶ Rigorous but not practical
- Numerical approach by using **gradient flow** [Lüscher '10]
- Application to **3D winding number** ($g : T^3 \rightarrow U(N)$)

$$W_3 \equiv \frac{1}{24\pi^2} \int_{T^3} \text{tr}(g^{-1}dg)^3 \in \mathbb{Z}$$

- ▶ cf. Shiozaki formula, $W_3^{\text{lat}} \in \mathbb{Z}$ ['24]
 - ▶ Analysis on gradient flow with “over-improved” action
 - ★ “Over-improvement” as instanton stabilization
[Tanizaki–Tomiya–Watanabe '24]
- Less computational burden, generalization to higher dim

Principal fiber bundle

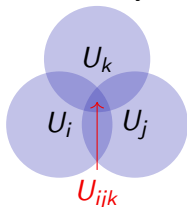
- Recall Dirac's discussion
 - Gauge fields cannot be defined globally in spacetime
 - Defined on each subspace: northern/southern hemisphere
- Manifold (spacetime) X ; open covering (patches) $\{U_i\}$
 - Gauge group G , gauge field a_i on U_i
 - Relation between a_i and a_j ?



Gauge transformation:

$$a_j = g_{ij}^{-1} a_i g_{ij} + g_{ij}^{-1} d g_{ij} \quad \text{at } U_{ij}$$

- Consistency condition for transition function g_{ij} :

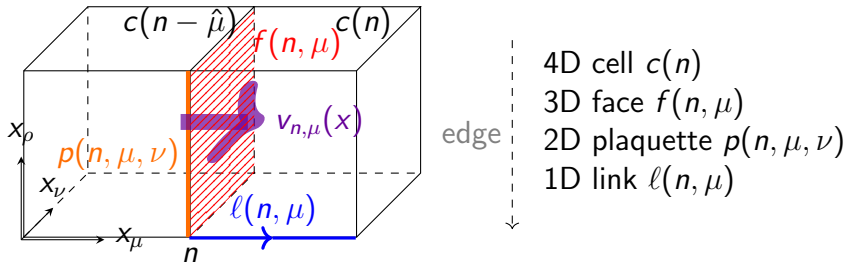


$$g_{ii} = \text{id}, \quad g_{ji} = g_{ij}^{-1}$$

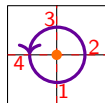
Cocycle condition: $g_{ij} g_{jk} g_{ki} = 1 \quad \text{at } U_{ijk}$

Bundle structure on lattice?

- Setup/strategy: Lattice Λ divides X into 4D hypercubes



- 1 Regard $\{c(n)\}$ as patches
- 2 Define transition function $v_{n,\mu}(x)$ at $f(n, \mu)$ from data as U_ℓ
 - Difficult to define it at $x \neq n$ s.t. cocycle condition is kept intact



$$v_1 v_2 v_3^{-1} v_4^{-1} = 1$$

at $x \in p(n)$

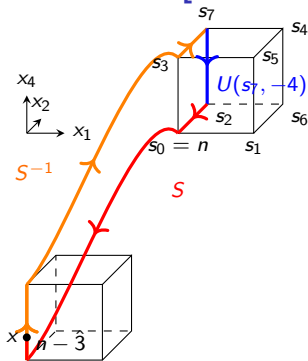
Construction of topo. sectors on lattice [Lüscher]

- Parallel transporter (interpolation):

$$U_\ell \rightarrow v_f(n) \rightarrow v_f(x) \Leftarrow \text{Cocycle}$$

$$v_{n,\mu}(n) = U(n - \hat{\mu}, \mu)$$

$$v_{n,\mu}(x) \equiv S_{n,\mu}^{n-\hat{\mu}}(x)^{-1} v_{n,\mu}(n) S_{n,\mu}^n(x)$$



- Topo. sectors on lattice so that $[Q \text{ in terms of } v_f(x)] \in \mathbb{Z}$

$$Q = \sum_{n \in \Lambda} q(n), \quad q(n) = -\frac{1}{24\pi^2} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} \left\{ 3 \int_{\rho_{n,\mu\nu}} d^2x \, \text{Tr} \left[(v_{n,\mu} \partial_\rho v_{n,\mu}^{-1}) (v_{n-\hat{\mu},\nu}^{-1} \partial_\sigma v_{n-\hat{\mu},\nu}) \right] \right. \\ \left. + \int_{f_{n,\mu}} d^3x \, \text{Tr} \left[(v_{n,\mu}^{-1} \partial_\nu v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\rho v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\sigma v_{n,\mu}) \right] \right\}$$

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$$w^n(x) = U_{n,4}^{y_4} U_{n+y_4\hat{4},3}^{y_3} U_{n+y_4\hat{4}+y_3\hat{3},2}^{y_2} U_{n+y_4\hat{4}+y_3\hat{3}+y_2\hat{2},1}^{y_1}$$

$$f_{n,\mu}^m(x_\gamma) = (u_{s_3 s_0}^m)^{y_\gamma} (u_{s_0 s_3}^m u_{s_3 s_7}^m u_{s_7 s_2}^m u_{s_2 s_0}^m)^{y_\gamma} u_{s_0 s_2}^m (u_{s_2 s_7}^m)^{y_\gamma},$$

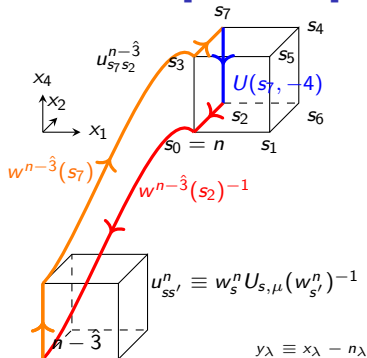
$$g_{n,\mu}^m(x_\gamma) = (u_{s_5 s_1}^m)^{y_\gamma} (u_{s_1 s_5}^m u_{s_5 s_4}^m u_{s_4 s_6}^m u_{s_6 s_1}^m)^{y_\gamma} u_{s_1 s_6}^m (u_{s_6 s_4}^m)^{y_\gamma},$$

$$h_{n,\mu}^m(x_\gamma) = (u_{s_3 s_0}^m)^{y_\gamma} (u_{s_0 s_3}^m u_{s_3 s_5}^m u_{s_5 s_1}^m u_{s_1 s_0}^m)^{y_\gamma} u_{s_0 s_1}^m (u_{s_1 s_5}^m)^{y_\gamma},$$

$$k_{n,\mu}^m(x_\gamma) = (u_{s_7 s_2}^m)^{y_\gamma} (u_{s_2 s_7}^m u_{s_7 s_4}^m u_{s_4 s_6}^m u_{s_6 s_2}^m)^{y_\gamma} u_{s_2 s_6}^m (u_{s_6 s_4}^m)^{y_\gamma},$$

$$l_{n,\mu}^m(x_\beta, x_\gamma) = [f_{n,\mu}^m(x_\gamma)^{-1}]^{y_\beta} [f_{n,\mu}^m(x_\gamma) k_{n,\mu}^m(x_\gamma) g_{n,\mu}^m(x_\gamma)^{-1} h_{n,\mu}^m(x_\gamma)^{-1}]^{y_\beta} h_{n,\mu}^m(x_\gamma) [g_{n,\mu}^m(x_\gamma)]^{y_\beta},$$

$$S_{n,\mu}^m(x_\alpha, x_\beta, x_\gamma) = (u_{s_0 s_3}^m)^{y_\gamma} [f_{n,\mu}^m(x_\gamma)]^{y_\beta} [l_{n,\mu}^m(x_\beta, x_\gamma)]^{y_\alpha}.$$

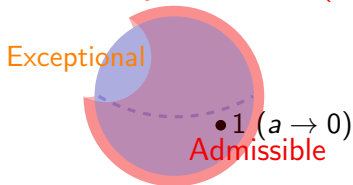


- Topo. sectors on lattice so that $[Q \text{ in terms of } v_f(x)] \in \mathbb{Z}$

$$Q = \sum_{n \in \Lambda} q(n), \quad q(n) = -\frac{1}{24\pi^2} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} \left\{ 3 \int_{\rho_{n,\mu\nu}} d^2x \, \text{Tr} \left[(v_{n,\mu} \partial_\rho v_{n,\mu}^{-1}) (v_{n-\hat{\mu},\nu}^{-1} \partial_\sigma v_{n-\hat{\mu},\nu}) \right] \right. \\ \left. + \int_{f_{n,\mu}} d^3x \, \text{Tr} \left[(v_{n,\mu}^{-1} \partial_\nu v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\rho v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\sigma v_{n,\mu}) \right] \right\}$$

Admissibility condition

- In general, admissibility = well-defined-ness of u^y ($0 \leq y \leq 1$)
 - ★ $U(1)$: $F_p = \frac{1}{i} \ln U_p$ for plaquette U_p
 - ★ $S_{n,\mu}^m(x)$ is written in terms of $(u_{ss'}^n)^y$ where u is a loop
 $n \rightarrow s \rightarrow s' \rightarrow n$
 - ▶ E.g., u^y is ill-defined at $u = -1$; ill-def regions separate **sectors**
- **Admissibility condition** $\text{tr}(1 - U_p) < \epsilon$ [Lüscher '84]



- ▶ Admissible lattice gauge fields:
well-defined conf space $\sim$ disk
 - ▶ Exceptional region
 - ★ Topological freezing
 - ★ Monopole as lattice artifact
 - ▶ Wilson action $S[U] = \sum_p \beta [\text{tr}(1 - U_p) + \text{c.c.}]$
 $\rightarrow$ action density should be sufficiently small.
- Under the admissibility condition, $Q \in \mathbb{Z}$; we observe topo. sectors even on lattice!

Numerical simulation of lattice gauge theory

- Interpolation techniques (not unique) provide
 - ▶ proof of integer-valued topological number,
 - ▶ but unuseful formula in numerical studies.
- Assurance of topology even on lattice
 - Some kind of smearing works as approximation.

Gradient flow [Lüscher 2010]

- **Gradient flow**: an evolution of fields and physical quantities along a fictitious time
- Gradient flow equation (gauge field)

$$\partial_t B_\mu(t, x) = -g_0^2 \frac{\delta S_{\text{YM}}}{\delta B_\mu} = D_\nu G_{\nu\mu}(t, x),$$

$$G_{\mu\nu}(t, x) = \partial_\mu B_\nu - \partial_\nu B_\mu + [B_\mu, B_\nu]$$

and the boundary condition

$$B_\mu(t=0, x) = A_\mu(x)$$

- t : mass dimension -2
- Diffusion equation with the diffusion length

$$x \sim \sqrt{8t}$$

Smoothing effect for the configuration as $t \rightarrow \text{large}$

Flowed composite operator [Lüscher–Weisz 2011]

- Correlation function of the flowed gauge field ($t_i > 0$)

$$\langle B_{\mu_1}(t_1, x_1) \dots B_{\mu_N}(t_N, x_N) \rangle$$

⇒ **UV finite** without the wave function renormalization

▶ Momentum integral $\int d^D p \rightarrow \int d^D p \exp(-tp^2)$

- Independence of the regularization

→ Probes of universal properties of theories

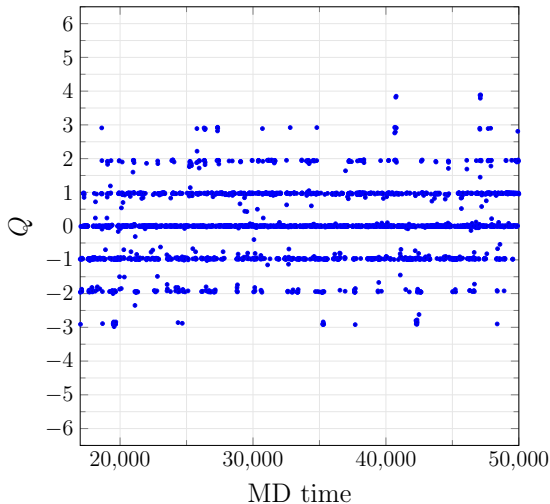
- Compatible with **gauge invariance**

- Many interesting applications to lattice gauge theory

- ▶ Topological charge
- ▶ Chiral condensation
- ▶ Energy-momentum tensor
- ▶ Non-perturbative gauge coupling

Gradient flowed topological charge

Q : naive discretization of $G \wedge G$, $\sqrt{8t} = 0.7L$



4D $SU(2)$, $\beta = 2.5$, $L = 12$ [Abe–OM–Suzuki '25]

Winding number on 3D

- Mapping $g: \mathcal{M} \rightarrow U(N)$, where $\mathcal{M}$ is a 3D closed manifold.
- Winding number

$$W_3 \equiv \frac{1}{24\pi^2} \int_{\mathcal{M}} \text{tr}(g^{-1}dg)^3 \in \mathbb{Z}$$

appears in various branches of physics.

- This is related to
 - ▶ 3D Chern–Simons theory/2nd Chern number on 4D
 - ▶ Class CII Hamiltonian system (chiral & PT symmetric)

$$H_{\mathbf{k}} = \begin{pmatrix} 0 & q_{\mathbf{k}} \\ q_{\mathbf{k}}^\dagger & 0 \end{pmatrix}, \quad q_{\mathbf{k}} = \mathbf{1} \otimes R_0(\mathbf{k}) + i\sigma_\mu \otimes R_\mu(\mathbf{k}).$$

$R_A(\mathbf{k})$: $n \times n$ real matrices. $g = UV^\dagger$ under SVD $q = U\Sigma V^\dagger$.

- In what follows, $\mathcal{M} = T^3$.

Shiozaki discrete formula of W_3

- $\exists B, \{U_j\}_j$ s.t. $W_3|_{U_j} = \frac{1}{2\pi} \int_{U_j} dB$ for each path U_j

- Diagonalization:

$$g = \gamma \Lambda \gamma^\dagger, \quad \gamma \in U(N), \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_N),$$

then, B depends on $\ln \Lambda$ ($\rightarrow \Lambda^{-1} d\Lambda$) [Gawedzki–Reis '02].

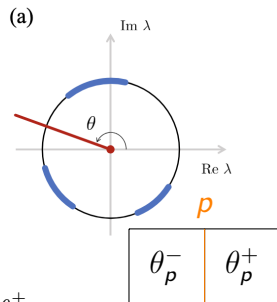
- Ambiguous due to branch of $\log$!
- For distribution of $\{\lambda_i\}_{\text{on } U_j}$, $\exists \theta$ s.t.
 $e^{i\theta} \neq \lambda_i$ if U_j is small enough.
- Well-defined $\log$: $\log_{\theta_j} \Lambda$ on U_j , where

$$\log_{\theta} z = \log |z| + \arg z,$$

$$\theta \leq \arg z < \theta + 2\pi.$$

- $W_3 = \frac{1}{2\pi} \sum_p \int_p (B_{\theta_p^-} - B_{\theta_p^+}) = \frac{1}{2\pi} \sum_p \oint_{\partial p} \alpha_{\theta_p^\pm}$

α : Berry connection between $\theta_p^-, \theta_p^+ \rightarrow$ discrete Berry phase...



Simple prescription of gradient flow (continuum)

- Postulate that

$$\partial_t g = g \partial_\mu (g^{-1} \partial_\mu g).$$

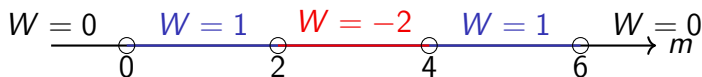
- (Positive semi-definite) Action is given by

$$S \equiv - \int_{T^3} \text{tr}(g^{-1} \partial_\mu g \cdot g^{-1} \partial_\mu g) \geq 0.$$

- One-parameter family of mappings ($T^3 \rightarrow SU(2)$):

$$g(\theta) \equiv (\xi_0 \mathbf{1} + i \xi_\mu \sigma_\mu) / \sqrt{\xi_0^2 + \xi_\mu^2},$$

$$\xi_0 \equiv m + \sum_{\mu=1}^3 (\cos \theta_\mu - 1), \quad \xi_\mu \equiv \sin \theta_\mu, \quad -\pi < \theta_\mu \leq \pi$$



Discretization of W_3 and flow action

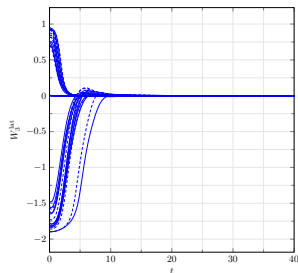
- Main component: $g^{-1}dg$
- Up to NLO, discrete expression is

$$\begin{aligned}
 H(x, \mu; \eta) &\equiv g(x)^{-1} \underbrace{\{g(x + \hat{\mu}) - g(x - \hat{\mu})\}}_{\text{LO}} \\
 &\quad - (\eta/6) \underbrace{[g(x + 2\hat{\mu}) - 2g(x + \hat{\mu}) + 2g(x - \hat{\mu}) - g(x - 2\hat{\mu})]}_{\text{NLO}} \} - \text{H.c.}
 \end{aligned}$$

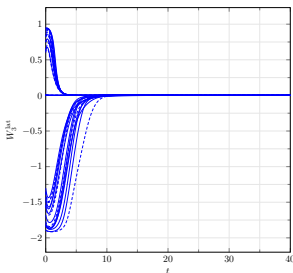
- $g^{-1}\partial_\mu g \rightarrow \frac{1}{4}H(x, \mu; 1) = \frac{1}{4}H(x, \mu)$ ($\eta = 1$: tree-level imprv)
- Then, winding number $\int (g^{-1}dg)^3$ and flow action become

$$\begin{aligned}
 W_3^{\text{lat}} &\equiv \frac{1}{4^3 \cdot 24\pi^2} \sum_x \sum_{\mu\nu\rho} \epsilon_{\mu\nu\rho} H(x, \mu) H(x, \nu) H(x, \rho) \\
 S^{\text{lat}}(\eta) &\equiv -\frac{1}{16} \sum_x \sum_\mu H(x, \mu; \eta)^2 \geq 0
 \end{aligned}$$

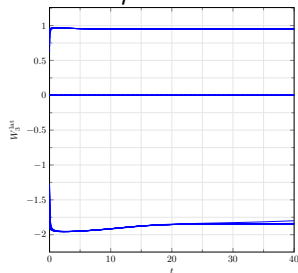
$W_3^{\text{lat}}(t)$ with random fluctuations, $L = 10$



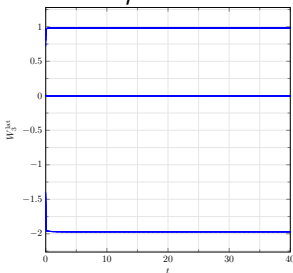
$\eta = 0$



$\eta = 1$

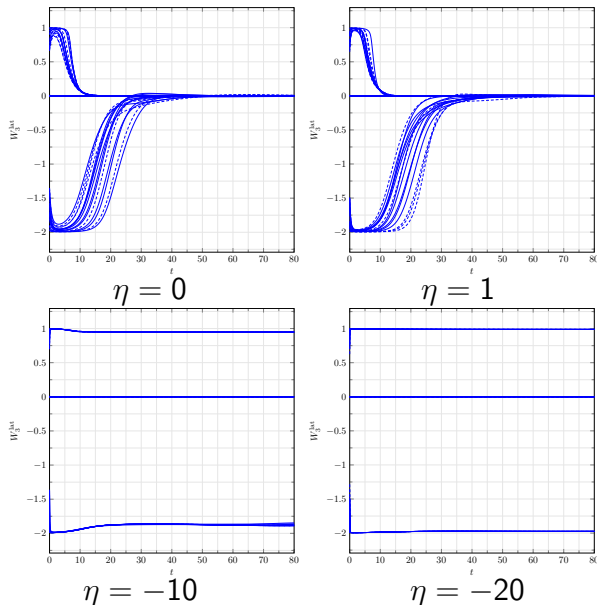


$\eta = -10$

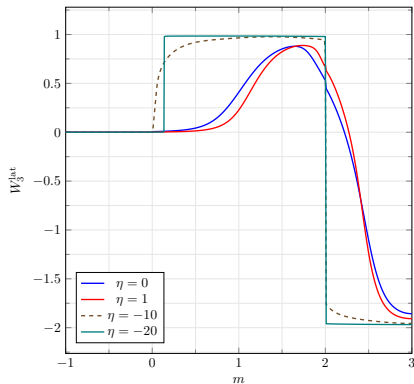


$\eta = -20$

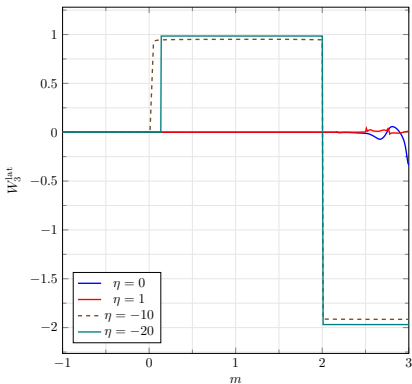
$W_3^{\text{lat}}(t)$ with random fluctuations, $L = 20$



$$W_3^{\text{lat}}(m), L = 10$$



$t = 2$

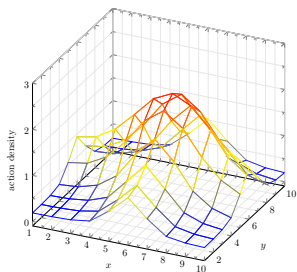


$t = 10$

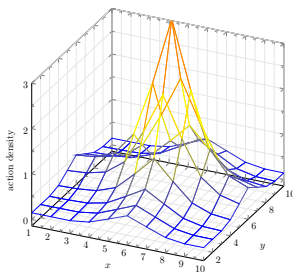
(De)Stabilization effect of gradient flow

- We observe that
 - ▶ $\eta = 0$ (no improvement), $\eta = 1$ (tree-level improvement):
 $W \rightarrow 0$
 - ▶ $\eta = -10, -20$ (“over-improved”): W nontrivial!!!
- What is happening?
- Similar to instanton stabilization of “over-improved” flow action in lattice gauge theory
 - ▶ Continuum: instanton at $t \rightarrow \infty$
 - ▶ Lattice: no instanton at $t \rightarrow \infty$ (“over-smearing”)
- Respect admissibility condition (small action density)

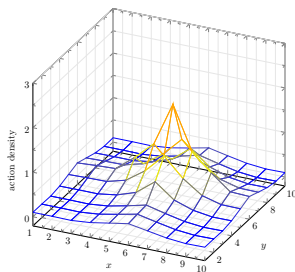
Action density along flow, $m = 1$, $\eta = 1$



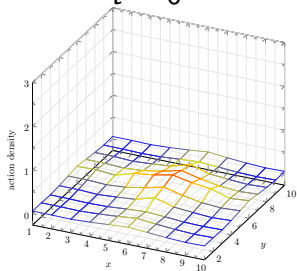
$t = 0$



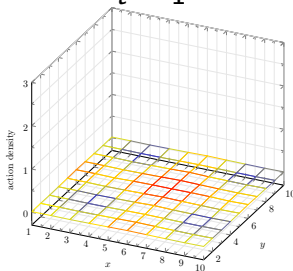
$t = 1$



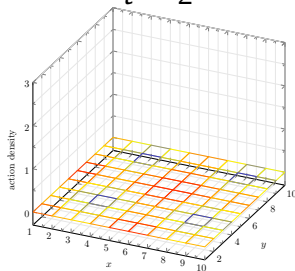
$t = 2$



$t = 3$

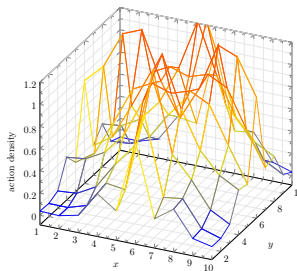


$t = 5$

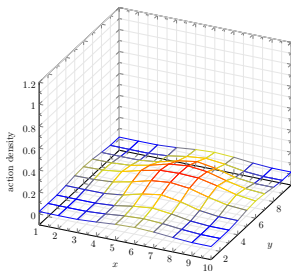


$t = 8$

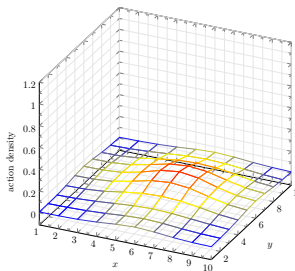
Action density along flow, $m = 1$, $\eta = -20$



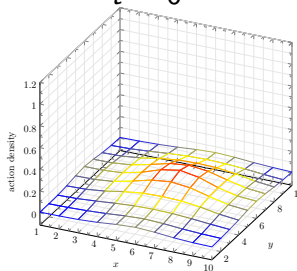
$t = 0$



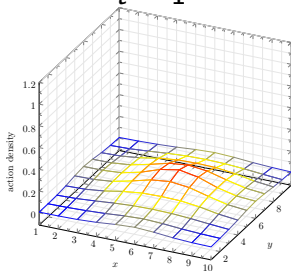
$t = 1$



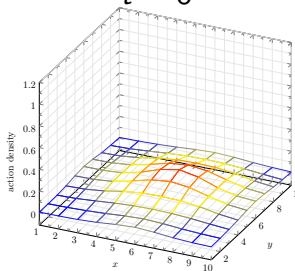
$t = 8$



$t = 15$



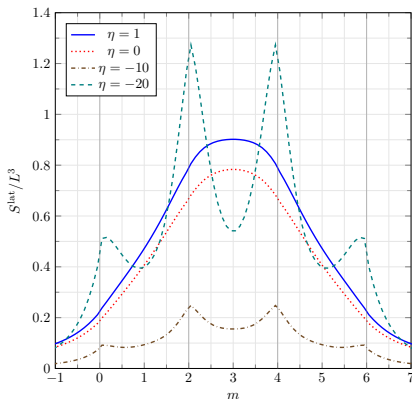
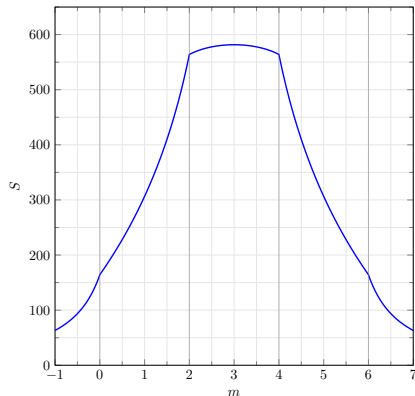
$t = 25$



$t = 40$

Lattice action of parameter m

- Boundaries between topological sectors
 - ▶ Cusp (left: continuum) $\rightarrow$ Barriers (right: lattice)



Summary

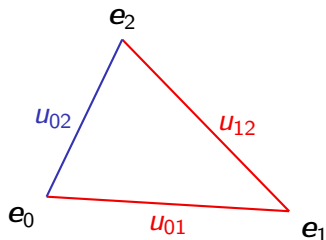
- We proposed simple and versatile numerical method
 - ▶ for computing winding number of mapping
 - ▶ approximated by discrete cubic lattice.
- Gradient flow with “over-improved” action which stabilizes lattice topological numbers:
 - 1 topological charge in 4D non-Abelian gauge theory, [Tanizaki–Tomiya–Watanabe, ...]
 - 2 winding number W_3 on 3D.
 - ▶ No interpolation, no diagonalization: low computational costs
 - ▶ Straightforwardly applied to other systems (class CII?)
- Principle: Higher-order improvement leads to enhancement of potential barrier at cusp (admissible along flow).
 - ▶ Some tunable parameters
 - ▶ Lorentz invariance (Skyrme term)

Backup: Definition of parallel transport function

- At first, let's illustrate an iterative method on nD $GL(p, \mathbb{C})$ **simplicial** lattice (triangle) [Phillips–Stone '86, '90]
- Let Δ^r be a r -simplex in X ; parallel transport function V as

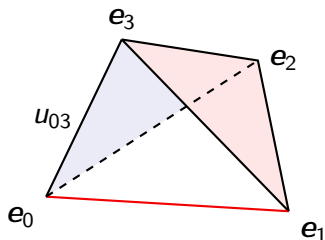
$$\begin{aligned}
 &V_{\langle 0 \dots r \rangle}(s_1, \dots, s_p = 1, \dots, s_{r-1}) \\
 &= V_{\langle 0 \dots p \rangle}(s_1, \dots, s_{p-1}) \\
 &\quad \cdot V_{\langle p \dots r \rangle}(s_{p+1}, \dots, s_{r-1}),
 \end{aligned}$$

$$\begin{aligned}
 &V_{\langle 0 \dots r \rangle}(s_1, \dots, s_p = 0, \dots, s_{r-1}) \\
 &= V_{\langle 0 \dots \hat{p} \dots r \rangle}(s_1, \dots, \hat{s}_p, \dots, s_{r-1})
 \end{aligned}$$



$$V(0) = u_{02}$$

$$V(1) = u_{01} u_{12}$$



$$V(0) = u_{02} u_{23}$$

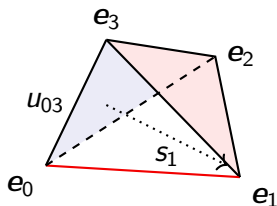
$$V(1) = u_{01} u_{12} u_{23}$$

...

Backup: Reconstruction of principal G -bundle

- Successive linear interpolation

$$V_{\langle 0 \dots r \rangle}(\dots, s_{r-1}) = (1 - s_{r-1}) V_{\langle 0 \dots r-2, r \rangle} + s_{r-1} V_{\langle 0 \dots, r-1 \rangle} u_{r-1, r}$$



- ▶ This is for $GL(p, \mathbb{C})$
- ▶ For other G , we need a projection or geodesic approach
 - ★ Recall $SU(3)$ projection in conf generation in lattice simulation

- (Math) Theorem

- ▶ Projection $\pi : E \rightarrow M$, section $H : M \rightarrow E$ s.t. $\pi \circ H = \text{id}_M$
- ▶ Definition of H

① if $\sigma = \langle i \rangle$, $H_\sigma : \mathbf{0} \mapsto \mathbf{e}_i$

② $H_{\langle 0 \dots r \rangle}(\dots, s_r) = (1 - s_r) H_{\langle 0 \dots r-1 \rangle} + s_r V_{\langle 0 \dots r \rangle} \mathbf{e}_r$

- ▶ Define $\Sigma_q = \{A | \text{rank}(A^1, \dots, A^{p-q+1}) \leq p - q\}$
- ▶ q th Chern class is represented by “intersection” of H_σ and Σ_q

Is $V_\sigma(\forall s)$ compatible with u_σ ?

- How different are $V_\sigma(\forall s)$ and u_σ ?

▶ $u_\sigma = u_{ij}$ where $\sigma = \langle i \dots j \rangle$

- For simplicity, e.g., in the $U(1)$ case,

- $\exists V(s)$ s.t. $\|V_\sigma - u_\sigma\| < \delta\varphi(K, n)$ for $\forall s$

▶ δ : a parameter

▶ K is maximum of operator norm

▶ n dimensions

▶ $\varphi(K, n)$ is determined by $\sim K^r, r \leq n$

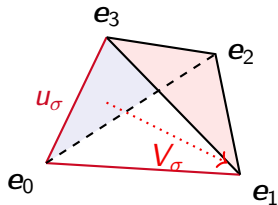
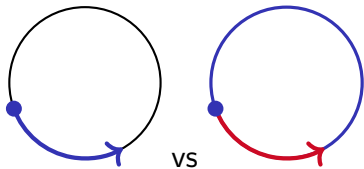
- Then, $\exists \epsilon$ s.t. if $\|u - u_0\| < \epsilon$

u determines the same bundle as u_0

- [NOTE] GL group makes the above statement simpler.

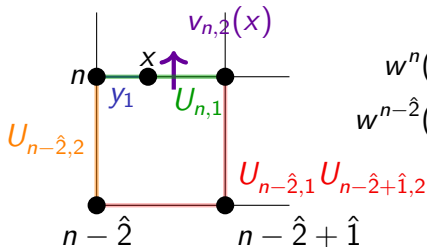
▶ These expressions were given by Phillips–Stone

▶ In general, *explicitly* gauge invariant forms are suitable



Exercise: Lattice 2D $U(1)$ gauge theory

- $v_{n,\mu}(x) = w^{n-\hat{\mu}}(x) w^n(x)^{-1}$ [Lüscher '98, Fujiwara et al. '00]



$$w^n(x) = U_{n,1}^{y_1}$$

$$w^{n-\hat{2}}(x) = [U_{n-\hat{2},1} U_{n-\hat{2}+\hat{1},2}]^{y_1} U_{n-\hat{2},2}^{1-y_1}$$

- Explicit expression of v :

$$\begin{aligned} v_{n,1}(x) &= U_{n-\hat{1},1} & v_{n,2}(x) &= U_{n-\hat{2},2} [U_{n-\hat{2},1} U_{n-\hat{2}+\hat{1},2} U_{n,1}^{-1} U_{n-\hat{2},2}^{-1}]^{y_1} \\ & & &= U_{n-\hat{2},2} \exp[iy_1 F_{12}(n - \hat{2})] \end{aligned}$$

- ▶ Field strength: $F_p \equiv \frac{1}{i} \ln U_p$ for $-\pi < F_p \leq \pi$
- ▶ (nD) To ensure Bianchi identity $dF_p = 0$, we should impose $\sup_p |F_p| < \epsilon$, $0 < \epsilon < \frac{\pi}{3} \rightarrow$ **Admissibility condition**